29 Oct '26

→ HW: For previous HW on drawing circles, for the last 4 circles, measure the Perimeter 5 times each with the rope. And then do this in excel

radius 'r'	P_1	P_1/r	P_2	P_2/r	P_3	P_3/r ...
5	32.1	?	31.4	?		

For each 'r' you get 5 P's.

Radius of a circle: The length of the line segment $\overline{PO}$ from any point on the circle P to the center of the circle $\overline{O}$.

Diameter of a circle: The length of the line segment $\overline{P_1P_2}$ from any point P_1 on the circle to another point P_2 on the circle such that $\overline{P_1P_2}$ passes through the center of the circle 'O'.

To find 'P_2' from P_1, we draw a line from P_1 to O and find where that line intersects the circle again.

Pi (π) : The ratio of any circle's perimeter (P) to the diameter of that circle is a constant called 'Pi'

$$\pi = \frac{P}{d} = \frac{P}{2r} \sim \frac{22}{7} = 3.14$$

π is an irrational number
(can't be written as a fraction or a decimal number with finite digits)

In our HW experiment, we were calculating

$$\frac{P}{r} = \frac{P}{2r} \times 2 = 2 \times \pi \sim 2 \times 3.14 = 6.28$$

we didn't get this exact value because our measurements have errors.

→ See AI notes and animations on how to measure π

→ HW: 4 different combinations of experiments and 5 measurements — (1) Paper strip and pin; Rolling ring to measure permeter (2) Protractor method or Book method

Find different objects to draw circles — rings, drinking glasses, flutes etc.

Each combination, repeat the experiment for 5 trials

For each type of experiment, take a picture.
Populate your results in excel.

Eg:

Perimeter type	Diameter type	object	Trial	p	d	p/d
Paper	Book	Hanging	1	?	?	?
"	"	"	2	?	?	?
			⋮			
			5	?	?	?
"	"	Drinking straws	1			
			2			
			⋮			
"	"	Flute	⋮			
"	Protractor		⋮			

2x2x3x5